Pharmacy Requirements

Feature 1 - User Account Management

Description: Allows users to create and manage their accounts within the pharmacy system. Users can register, log in, and update their personal information. The system validates all input data and securely stores user credentials.

Administrative users can manage user accounts through an admin dashboard, where they can search users and grant administrative privileges.

F1.1:- User login

Description: System shall display a login form accessible from the home page, containing 2 fields labeled “E-mail” and “Password”, a button labeled “Login” and, at the bottom of the window, a label containing “Don’t have an account? Register one now!”, which, when clicked will take the user to the registration form.

Validation: To successfully log in, a user has to provide:

- An e-mail already registered with an account, of valid e-mail form (e.g. “x@domain.com”)
- an e-mail may only have one account linked to it
- The password associated with their account
- Their account has to not be disabled
- If validation fails, the system shall display an error message, containing the invalid field

F1.2:- Account registration

Description: System shall display a registration form, accessible from the login form, from the label “Don’t have an account? Register one now!”, which will contain 5 fields labeled: "E-mail", "Phone Number (optional)”, ”Username (optional)”, “Password” and “Confirm Password” and a button labeled “Register Account”.

The system shall store user passwords in encrypted form in the database.

Validation: To successfully create the account a user has to provide

- An e-mail not already linked with an account, if the mail already has an account linked to it, a message will be displayed saying ”E-mail already in use”
- A password which follows the requirements:
  -must be minimum 8 characters long
  -must contain at least an uppercase letter
  -must contain at least a lowercase letter
  -must contain at least one digit
  -must contain at least one special character from the list: {!,@,#,%,^,*}
  -must not contain characters not from the english alphabet (upper and lower case), from the special character list or digits

- The password, repeated in the “Confirm Password” field, if the 2 password field don’t match, an error message is displayed saying: “Passwords don’t match”
- A phone number, containing only digits, which is an optional input
- Username, text containing only letters from the english alphabet (upper and lower case) and/or the character “_”, optional input
- If validation fails, the system shall display an error message indicating the invalid field

F1.3:- Profile management

Description: System shall allow logged-in users to view their account information in a separate form, accessible from a button on the top bar, which also has the text next to it “Hello, <username>!”. The form contains:

- A button with a clipboard icon, called “Order History”, which would take the user to their order history page
- The e-mail linked to the account
- Phone number linked, empty if none was provided
- Username field (empty on creation, but would allow the user to change how the application refers to them by modifying it)
- Password field, with a label underneath: “Change Password”
- Button labeled “Save changes”
- Button labeled “Cancel”

The user is allowed to modify the “Username” field (including leaving it empty, in which case the local part of the e-mail (the part before “@”) will be used anywhere the username is supposed to) and the phone number section, which has the same restrictions as when creating the account. Clicking cancel would exit the form without modifying any user information, while clicking “Save changes” would try to modify the changed fields. If validation fails an error message is displayed in a pop-up window saying which field is invalid. If only one field is invalid, but multiple were changed, none of the changes are accepted. If validation is successful, a message will be displayed saying “Changes saved successfully”.

When clicking the “Change Password”, a new form will be opened asking the user to input their old password, a new one and to retype the new password. If the 2 new passwords don’t match or the old password is incorrect or the new password does not comply with the password rules, an error message will be displayed and no changes will be made. There will also be a button labeled “Change Password”. If everything is successful, the form will close and a message saying “Changed password successfully” will be displayed.

All changes must be stored permanently in the database.

Administrators shall have access to a “Dashboard”, not visible to clients, accessible from the home page, which when clicked, will show a small menu where all the admin related feature buttons will be.

F1.4:- Admin Account modification

Description: Admins shall be able to view and modify certain user information of every created account, through a separate window, accessible from the admin dashboard.

The form contains:

- Search bar which allows admins to filter the users shown through userId, username or e-mail, using certain prefixes in the search (“id:”, “username:”, “mail:”)
- The user accounts will be shown as info cards with the information: e-mail, username (blank if empty), phone number and a checkbox with the label “Admin”
- Admin accounts will have the info cards visually highlighted and the checked box greyed and unmodifiable
- An option to disable the account, only shown for client account

Validation:

- For client accounts, admins can check the admin box, which will make a warning appear, saying that this action cannot be undone
- If “Proceed” is selected, the selected user becomes an Admin and gets access to all the admin features
- If the option to disable the account is selected, a warning will appear
- If the admin decides to proceed, the user account will be marked: Disabled and will no longer be accessible by the user
- The user cannot create another account with the same e-mail

Feature 2 - Order management

Description: Allows users to place, view, modify, cancel and resubmit pharmacy orders. The system maintains order status and expiration information and validates all operations against inventory availability and medicine expiration dates.

Administrators can view all orders, complete pending orders, and undo completed orders through an order management interface.

F2.1:- Order placement

Description: When a user proceeds to checkout from the basket, the system shall display a checkout page where the user can review their selected items and choose a pickup date.

The page shall contain:

- List of medicines from the order, sorted by when they were added to the basket
- The quantities of each medicine
- Total price
- A calendar menu where the user can select a pickup date
- A button labeled “Place Order”
- A button labeled “Cancel”

The system shall automatically suggest a pickup date equal to the next day. Users may select any later date using the date selector.

Validation: To successfully place an order:

- The pickup date must be at least one day after the current day
- All the medicines have an expiration date later than the pickup date
- If the selected pickup is invalid, the system must display the message: “Pickup date must be at least one day in the future”

If validation is successful:

- The order shall be stored in the database
- The order shall be assigned status of "Incomplete"
- No medicines should be removed from the inventory
- Order marked as Not Expired
- Order shall receive an expiration date of 7 days after selected pickup date

F2.2:- Order History

Description: The system shall allow users to view their order history in a separate page, accessible from their account information tab.

The order history page shall contain the orders linked to the user's account, displayed as info cards, each containing:

- OrderID to give to the pharmacist
- Order creation date
- Pickup date
- Order status (Complete/Incomplete)
- Order expiration date
- A button labeled "Cancel Order" for incomplete orders

Order statuses can be:

- Incomplete, no medicines/items have been taken out of the inventory and the order has to be completed by an admin
- Complete, after being approved by an admin

Orders which are expired shall be visually highlighted in the order history list and in the admin's order page.

F2.3:- Modification of Incomplete Orders

Description: Users shall be able to modify Incomplete orders which have not expired, from the Order History page.

When clicked on an order, a list of all medicines with respective quantities will be displayed in a pop-up window, which will contain:

- An option to decrease/ increase the quantity, if there are not enough items in the inventory to increase, an error message will be displayed
- An option to remove an item entirely
- A button labeled "Add item", which would allow the user to search for other items and add them to the order list, through a separate window, containing a search bar for the name and a list of the resulting medicines through partial matching
- A button labeled "Change Pickup Date", which would allow the user to select another date for pickup in the future. En error message will be displayed if the new selected pickup date is later than the expiration date of any old/newly added medicines
- A button labeled "Save changes"
- A button to "Cancel", which would leave the order as it was

Validation: If the “Save changes” button is pressed and there are no issues regarding the new contents, the window will close, the order information will be modified permanently and a message with success will appear.

F2.4:- Order cancellation and Expiration

Description: The user shall be able to cancel any Incomplete order from the Order History tab, by clicking the “Cancel Order” button on the desired order. There will also be an option in the order menu to show expired orders using a checkbox.

When pressed:

- A pop-up window, asking for confirmation will appear with an option to cancel and one to proceed
- After confirming, the order shall be marked as “Expired” and will no longer be editable
- If the option to show them is ticked, expired orders will be shown in the list, visually highlighted

Orders are automatically marked as “Expired” after their expiration date passes.

Expired orders shall have the option to “Cancel” replaced with the option to “Resubmit”, which will open a new checkout window with the same items where the user will have to pick a new pickup date.

Validation: Same validation rules apply as for F2.1. If validation is successful:

- New order with new orderId shall be created
- Original expired order shall remain unchanged

F2.5:- Admins and orders

Description: Admin users shall have access to an Order Management Page, accessible from the home page, different from their own Order History, where they will see a list of all orders, initially empty, but with the option to show all orders by inputted userId and/or orderId, a checkbox for showing only incomplete orders and one for showing only orders that have not expired.

The page shall display the following information for each order, grouped as info cards:

- Order ID
- User ID
- Order status, which shall be colored differently depending on completeness
- Pickup date
- Expiration status, expired orders shall be visually highlighted
- Expiration date

Selecting an order shall display the order details, similar to users seeing their own orders in the history.

Admins shall have the following options in the order management page, for each order:

1. Undo completed order
    a. Visible only for complete orders
    b. Selecting this will try to add the medicines back into the inventory
    c. If none were expired, success

    d. If there are any that were expired, the system will show an error message, showing a list of the expired items with respective quantities with options to cancel the undo and to proceed
    e. Proceeding will add only the items that have an expiration date later than the current date, the others being thrown away for safety
    f. Success will prompt a message, the order will be marked as incomplete and expired, if it had any expired items
2. Complete order
    a. Visible only for incomplete orders
    b. Selecting this will attempt to complete the order, by checking to see if the medicines that are in the order's list are still available in the desired quantity, if the same medicines are in stock, but with different box-quantities or expiration date, the system will consider them as equivalent and try to complete the order replacing the unavailable medicines, trying to match the quantities desired
    c. Failure to meet this requirement will result in an error message, telling the admin that the inventory is insufficient, with the option to close the pop-up
    d. To fix order failures, the admin will have to either modify the order, removing the unavailable items, or adding in stock items that were desired.
    e. On success, the order shall be marked as completed, the inventory shall be updated accordingly and a message announcing success shall be shown

F2.6:- Basket

Description:

- The top bar of the page shall have a button with a cart icon that when pressed brings the user to the basket's page; the icon should display a number on it signalling the number of distinct products placed inside the basket in that moment
- The basket is stored locally in memory, being emptied if the user exits the application
- The basket detail page has a list view with all the products that the user wants to buy, displaying name, producer, price and quantity bought; price should be calculated from the product's initial price and the discounts that are applied to that product, queried from the database, while quantity should be an input field for integers that can be modified either by typing another integer or using two buttons next to it to increment/decrement it next to the list view there should be a text displaying the price of all the products in the basket (in a big red text), and if there are any active discounts on any of the products, it will display the original cumulative price in small gray strikethrough text, while the discounted price will be displayed in a big red text
- The basket detail page a button with the name "Checkout" that brings the user to the part specified in F2.1 (picking pickup date and creating order)
- The user shall be able to access any product's detail page from the basket
- The system shall check the contents of the basket every time the user enters the basket's page and have a list of 3-5 products below the basket's list view that recommends over-the-counter medication that was bought in the past with the heading "Because you bought these in the past"; clicking them would bring you to the

product's detail page; it should also recommend over-the-counter products that are similar to the ones bought in the past based on containing the same active substances or being in the same category
- The system shall check for every over-the-counter medication in the basket if there are available medications that have the same active substances, their concentration per pill is a divisor of the concentration of the medication in the basket and their price is lower; if any medicine exists that completes these requirements, then another list view will appear below the basket list view (and other recommendations based on order history) with the title "Cheaper alternative for <insert name of the product that can be replaced>" which contains the medication that completed the aforementioned requirements
- in the basket's list view every product would have a button next to it with a trash can icon, that deletes the appropriate product from the basket; the user could also delete a product from the basket by modifying its quantity to 0; the user shall be able to undo this action, by clicking an "Undo delete" button at the top of the basket list view, which will bring back the last deleted product with the last quantity before deletion (so undo should hold the product and a quantity with it); the user shall be able to use undo multiple times; undo history is stored locally, erased on application exit

Validation:
- If the undo history is empty or the user has arrived at the beginning of the undo history, pressing the undo button will result in a message pop-up that says "no more deleted products

F2.7:- Add items from prescription (cross-team: MysteryInc)

Description: In the basket page, users have a field to input a prescription ID, together with a button called "Add Products". This will add the products from that description into the basket of the user, or increment the ones already in the basket.

For products in the prescription that are not in stock, add one that is equivalent in terms of concentration of the active substances, with the same number of pills in the package, that is cheapest.

Validation: If the inputted prescription id doesn't exist, write with red the text "invalid prescription ID" below the textbox.

If there aren't any with the same number of pills, then add more of the products that are equivalent but have less pills, such that its still cheapest and the total number of pills for that product is greater than or equal to the amount on the prescription (for example, add two packs of 20 pills instead of one pack of 30 pills).

If there aren't any drugs with the same concentration of active substances, get more packs of ones that have, approximately (with an error), the same concentration when added, and the total number of pills will be double the number of pills written in the prescription. If even then, there aren't any items that can match it, just skip it and the pharmacist will tell you that physically.

If the prescription also has a **discount** on it (for example people with some insurance), then take that off from the whole price. This will be validated by the pharmacist with the physical prescription

Feature 3 - Pharmacy Management & QoL

Description: Allows users and admins to manage the pharmacy in different ways. Admins can update the inventory, use statistics to their benefit, the application is sending notifications to users on different events, and the users can file a complaint to, in turn, manage the pharmacy themselves, free labor. Admins are dictators who can just delete your complaint because

F3.1:- Updating Pharmacy Inventory

Description: The application will display a page with multiple tabs, each containing a table. Below each table in each tab, there will be 1 button: remove (removes the selected entity from the table and database); and 2 buttons that open 2 sub-windows for adding and updating an entity: add and update (which update the table and the database). There will be a few labels with textboxes for the latter 2; these will concern the fields of the entities in the respective table, to add or update. On add it adds a new entity with the given values. On update it updates an existing entity given by its *ID*. Admins can also have an option to toggle to view only **expired items** in the Items table, to remove them easier. Also, admins can *search* for items by their name in the table. This will be done from a search box that is above the table.

1. The first table is the ***Items*** table.
    a. On add, you have X fields: the name, the expiration date, the category, the number of pills in the package, the quantity
    b. On update, you have X fields: the ID, the name, the expiration date, the category, the number of pills in the package, the quantity
    c. On remove, it also removes the item from all orders
2. The second table is the *Substances* table:
    a. On add, you have 3 fields: the name and the intake limit
    b. On update, you have 3 fields: the same as for add

    Validation:
    - On update, it updates only if the ID already exists in the table, otherwise it does nothing.
    - If in add or update any of the fields is empty and mandatory, it highlights them red, and displays an error “mandatory fields are empty”.
    - If the fields do not meet the required *format* (for example, the quantity being an integer for Items).
    - Some fields are not mandatory and they have a default value
        - Mandatory fields will be: for Items, everything except the category and the quantity, them defaulting to Medicine, respectively 1; for Substances: everything

F3.2:- Notifications

Description: Every user has a dropdown menu in the upper right corner that opens up, where notifications show up. A notification is one item in the dropdown menu that has some text, a title and some content. The title is concerning the category of notification, and the body what entity(es) that notification is about. The notifications are:

- **Product in stock**: this notification is received by all users and it happens when a user has subscribed to an item to be notified every time the item is out of stock. This is a button on that product's page and you need to be logged in to subscribe. The text body of this notification is the title of the product: name, number of pills, concentration and producer. Clicking it brings the user to the product page.
- **Product expired:** this notification is received only by admins, and it concerns a Product that is expired and that they have to remove from the database and the physical pharmacy. The text body is of the form "Product: ID expired. Please remove it". Clicking it brings the admin to the dashboard with products.
- **Discounts (we're not desperate I swear)!!:** this notification is received by all users who opted in for discounts. The text body is the amount of discount, for example "20% discount on 'Product name'!!". Clicking it brings you to the product page.
- **Complaint**: This notification is received when an admin answers your complaint or closes it. Clicking it opens up the complaint thread. It is also received by admins when a user answers a complaint.
- **Review**: This notification is received by admins when someone flags/reports a review to be checked by an admin.

Validation:

- **Product in stock**: if the product's quantity > 0, then we give the notification to all users who have it turned on for that product
- **Product expired:** if the product's expiration date is today or it passed for a product ID, throw that batch (a specific quantity that expires on that date that the pharmacy has in their inventory. An item (identified uniquely by ID) can have multiple batches, expiring at their respective expiration dates) out.
- **Discounts:** If an item has a discount (generdal percentage one, or quantity bought based discount (buy 4, get a 20% discount on all for example)), notify all users who opted in to receive discount notifications
- **Complaint:** notify only the user that submitted the complaint, not others. Also notify the admin when a user answers to a complaint that they are the moderator of
- **Review:** This notification is only received by admins. Anyone can flag a review in the product catalogue

F3.3:- Statistics for admins

Description: Admins are able to see in a list view in their dashboard Statistics, which will be a page in their dashboard, with multiple tabs.

- The first tab will have **popularity statistics**, which will display in order the top 30 products sold in the last month, this helps the pharmacists know what supply to order next. Besides this, we will show another list side by side with this one that uses the substances and concentration in the top 20 products sold to predict the most popular products of the next month
- The second tab will have the most **loyal clients**, which will display the top 10 clients (by money spent) in the last month. These users will have a general personalised discount of 20% on their order until they fall out of the top 10
- The third tab uses the bought products in the last month to find out the most **popular substances** (based on how many popular products contain it), shown in a list with name and details, and then predicts the most popular substances for the next month (the list views will be shown side by side, and in each there will be 30 active substances), and it helps the pharmacist know what supply to order for next month.

Validation: The algorithm used to predict statistics has to be pretty accurate. To successfully predict, we need to have the top 30 products sold, respectively top 30 active substances sold, i.e. if there aren't 30 products, or, respectively, active substances in the top, because there weren't enough products bought that month, file for bankruptcy, or just don't display statistics and yolo it

F3.4:- Moderation

Description: Admins have the ability to moderate **reviews** for items and delete them if they don't meet the rules of a review (be polite at least). Admins can close or answer complaints

Validation: To successfully close a **complaint,** admins need to provide a justified reason (e.g. "Case solved") when they are closing it. If a user is trolling, an admin can close a complaint without responding to it. An admin need not to justify the reason for deleting a review.

Feature 4 - Special Offers and Prizes

Description: Allows admins to manage (add, delete, update) discounts and special offers. Allows users to benefit from discounts and special offers, as well as accumulate coupons and loyalty points which can be used at checkout.

F4.1:- Discount Management (Admin)

Description: The application shall display a page with two tabs: Discounts and Bulk Discounts.

The Discounts tab:

The Discounts tab shall contain a form and a table. The form contains fields for MedicineID, BoxSize and DiscountPercent and buttons Add, Update and Delete. The table contains columns for MedicineID, MedicineName, BoxSize and DiscountPercent. Each row in the

table represents an active discount. The table can be sorted by MedicineID or DiscountPercent, ascending or descending.

Selecting a discount from the table:

Clicking on a row in the table will transfer the corresponding information to the form. Thus the form can be filled out either manually or by clicking on an entry in the table.

Adding a discount:

Upon filling out the form and pressing Add, an entry will be added to the table with the corresponding information and the database entry of the product with the matching MedicineID and BoxSize will be updated with the DiscountPercent introduced.

Updating a discount:

Upon filling out the form and pressing Update, both the table and the database entries with the corresponding MedicineID and BoxSize will be updated with the DiscountPercent introduced.

Deleting a discount:

Upon filling out the MedicineID and BoxSize fields in the form and pressing Delete, the corresponding table entry will be deleted and the database entry will be updated with the DiscountPercent null.

Recommending products for discounts:

The page should have a list view below the discounts view, that is loaded with 5-7 items displaying its id, name and price that the system suggests products that should be discounted based on whether they are about to expire after a week, have sold the fewest or have sold the most.

Validation:

- To delete a discount, the MedicineID and BoxSize fields of the form must be filled out. To add or update one, every field must be filled out.
- To add a discount, one must not already exist for the product with the specified MedicineID and BoxSize. To update or delete a discount, it must exist.
- The product with the matching MedicineID and BoxSize must exist for all operations.
- To add or update a discount, DiscountPercent must be an integer between 1 and 100.

If any of these conditions are not respected, the operation fails and an error message is displayed, indicating the problem.

F4.2:- Special Offer Management (Admin)

Description: Special Offers allow admins to group together multiple products at a fixed price.

The application shall display a page containing a form and a table. The form contains fields for OfferID, OfferName and Price and buttons Add, Remove and Update. The table

contains Columns for OfferID, OfferName and Price. The table can be ordered by any field, ascending or descending.

Selecting an offer from the table:

Clicking on a row in the table will transfer the corresponding information to the form. Thus the form can be filled out either manually or by clicking on an entry in the table. Additionally, a detail view of the selected offer will be displayed in a separate window (only one such window can be open at a time, if one already exists it will be updated when a new offer is selected).

Adding an offer:

Upon filling out the form and pressing Add, entries will be added to the table and the database with a newly generated offerID and the corresponding OfferName and Price.

Updating an offer:

Upon filling out the form and pressing Update, both the table and the database entries with the corresponding OfferID will be updated with the OfferName and Price introduced.

Deleting an offer:

Upon filling out the OfferID field in the form and pressing Delete, the corresponding table and database entries will be deleted.

Validation:

- To add or update an offer, all fields must be filled out. To delete one, only the OfferID field is necessary.
- To add or update an offer, the Price must be a float greater than 0 and the OfferName must be between 1 and 100 unicode characters.
- To update or delete an offer, an offer with the specified OfferID must exist.

If any of these conditions are not respected, the operation fails and an error message is displayed, indicating the problem.

The Detail View:

The detail view shall contain a label, a table, and a form. The label shall display the total price of every item in the offer. The table has columns for MedicineID, MedicineName, BoxSize, Quantity and Price (price of the product * quantity). It can be sorted based on MedicineID, Quantity and Price, ascending and descending. Each row in the table represents a product in the special offer selected in the main window. The form has fields for MedicineID, BoxSize and Quantity and buttons Add, Update and Delete.

Selecting a product from the table:

Clicking on a row in the table will transfer the corresponding information to the form. Thus the form can be filled out either manually or by clicking on an entry in the table.

Adding a product:

Upon filling out the form and pressing Add, an entry will be added to the table with the corresponding information and an association will be created between the offer and the product (with the same MedicineID and BoxSize) in the database, also containing the product quantity in the offer. The label will be updated to reflect the new total price of the offer.

Updating a product:

Upon filling out the form and pressing Update, the table entry with the corresponding MedicineID and BoxSize as well as the database association will be updated with the specified Quantity. The label will be updated to reflect the new total price of the offer.

Deleting a product:

Upon filling out the MedicineID and BoxSize fields in the form and pressing Delete, the corresponding table entry and database association will be deleted. The label will be updated to reflect the new total price of the offer.

Validation:

- To add or update a product in an offer, all fields must be filled out. To delete one, the Quantity field isn't necessary.
- To add, update or delete products in an offer, the selected offer must exist at the time of the operation (an offer could be selected and then deleted in the main window for example, making this validation necessary).
- To add, update or delete products in an offer, the product with the specified MedicineID and BoxSize must exist.
- To add or update products in an offer, the specified Quantity must be greater than 0.

F4.3:- How users interact with discounts

Description: Users will be able to see the price of an item together with the discounted price. The old price is striked through, and the new price is shown to its right with another color. The discount percentage, "-X%" is shown to the left of the striked through price. The users can also receive a **bulk discount**, based on how many of the same item they buy. The rule is that they receive a 10% discount on those items for every 5 items they add, stopping at 25 items.

Offers are shown in the store as normal items with a different category, and they contain a few products that will be sold at an overall discounted price. When a user clicks an offer to open it into a detailed view, they see the items in a list, together with the details: item image, item title, item price (real price, because the discount is to the whole bundle). At the end of the vertical list, there will be a button "Add bundle to basket"

Validation: A user is only able to see a discounted item with the striked through price if it is discounted, not otherwise. Bulk discounts are automatically applied to the basket and written in the "Discounts" section of the basket, right above the "Place order" button, for example "-2$ (bulk discount)". Offers (item bundles) are shown with the price striked through, and the discounted one (basically the bundle overall discount is applied) is shown near it, that's the real price of the bundle.

F4.4:- Loyalty system

Description: The system shall give loyalty points to every user after an order has been completed by the admin, which would be equal to the price of the order multiplied by 100. These loyalty points would accumulate after every completed order, and if it reaches a certain limit (for ex. 80000), then the user would receive a 5% discount on their next order. The user should be able to choose if they want to apply the discount or not, using up the loyalty points or not yet. Next tiers would be at 150000 for 10% and 200000 for 15%, with no more tiers. The user will indicate on the basket page whether they want to use up the loyalty points with a checkbox.

Validation: The user shall be able to view their collected loyalty points on the account info, and on the basket page they should be notified about the loyalty discount (if there are any in effect), displaying the percentage and the amount subtracted from the original order price. If the checkbox is unchecked, then the text with the percentage and subtracted price will disappear, with the final price of the order recalculated and its text updated with the removed discount. After applying the loyalty discount, the points should be changed to 0.

F4.5:- Mystery boxes

Description: When the user places an order, the order will also contain another item, if the user spends above 100 RON. This item will be titled “Mystery box” and the price of 0$. The mystery boxes are actually an offer (bundle) of items, that is randomly selected from the pharmacy selection. This is safe to do because a bundle doesn’t have prescription medications.

Validation: The chance for a user to receive a mystery box is 67%, and the order has to be above 100 RON. The user is able to refuse the mystery box at the pharmacy when they receive their order. If there are no bundles present in the database, a mystery box is just a bonus item that is not a prescription item from your current order.

Feature 5 - Product Catalogue

Description: The product catalogue is a functionality that deals with the showing of the products to the users and all the interactions that the user has with the items. This contains the Product Catalogue page but also the Home page.

F5.1: - Product Listing

Description: The product catalogue page shall display the products using a card-based layout. Each product card shall contain:

- An image
- The name of the product
- Product price in lei

- If an item is discounted it will show the new price and by how much it is discounted in percentages
- Average rating(5 star system with half star increments)
- Stock availability indicator(described in validation)
- A button labeled “Add to Basket”
- If the Stock is 0 the “Add to Basket” button shall be replaced with a button labeled “Set stock alert”. The stock alert button shall allow users to subscribe to an item and when an item is in stock, the subscribed users shall be notified (This is discussed more in pharmacy management)

Products shall be paginated, allowing users to navigate through multiple pages of items.
Products shall be ordered by their internal product IDs by default.

Validation:

1. Products with stock smaller than 10 shall be labeled with the text “Only **x** in stock”, “**x**” being the numbers of items that are in stock.
2. Products with stock equal to 0 shall be labeled with the text “Out of stock”.
3. The systems checks if an item had a discount percentage > 0 and if this is the case it is treated as a discounted item

F5.2:- Product Search

Description: The product catalogue page shall display a search bar allowing users to search products by name. The search shall return products whose name matches or partially matches the entered query.

Validation:

1. Search shall be case-insensitive.
2. If no results are found, the system shall display the message in gray: “No products found.”

F5.3: - Product Filtering

Description: On the left side of the page there will be a sidebar that includes some checkboxes with all the available filters. The filter name is the text that appears above the selectable options of every checkbox.

Available filters are:

- Product category: the options are all the categories available
- Price range: the options are: 0 - 49 lei, 50 - 100 lei, 100 - 199 lei, 200 - 499 lei and 500+ lei
- Stock availability: the options are: in stock, low stock (< 10 products in stock)
- Discounted products: the options are discounted products and non-discounted products this is the only category where you can only select one option
- Active substances: the options are all the active substances from all the items available

Below all the filters there is a button labeled “Select filters” which filters all the products based on the selected options.

Validation:

1. If no option is selected from a filter then the system shall display the results as if all the possible options for that filter were selected.

2. Category filter
    - All the items that belong to one of the categories selected shall be displayed
3. Price range filter
    - The system checks if the item price is >= first number in the range (in 0 - 49 lei range that would be 0) and the item price <= last number in the range (in 0 - 49 lei range that would be 49) and it displays all the items in this range
4. Stock availability filter
    - When the option selected is “in stock” the system displays all the items that have the stock > 0
    - When the option selected is “low stock” the system displays all the items that have the stock > 0 and < 10
5. Discounted products filter
    - When the option selected is “discounted products” the system displays the items that have the discounted field open
    - When the option selected is “non-discounted products” the system displays the items that have only a current price
6. Active substances filter
    - All the items that have one of the active substances selected shall be displayed
7. If no products match the selected filters the system shall display the message: “No products match the selected filters”.

F5.4:- Product Sorting

Description: In the same sidebar there will be a dropdown menu that will have multiple sorting criterias. The user is able to select one of them at a time. (the default sorting is by item id) These criterias are price, rating, and newest products. Below this dropbox there shall be another one that has the criterias Ascending/Descending (the default option being Ascending).

Validation:

1. Only one criteria can be active at a time
2. If no sorting option is selected, products shall be displayed in the default order defined above.

F5.5: - Product Details Page

Description: When a user selects a product card, the system shall open a product details page containing:

- A large product image in the left part of the page
- Product name besides the image
- Price in lei below the product name. If the product has a discount, the original price shall be displayed in strikethrough format and beside it the discounted price and by what percentage. For example: “~~100~~ 25 lei 75% off”
- Average product rating below name (they look like 5 star ratings with half increments)

- Stock quantity if it is < 10 otherwise display the text “In stock” or “Out of stock” depending of the stock number (the system shall display it below rating)
- Quantity selector below product rating(an editable textbox with the quantity number)
- Add to basket button besides quantity selector
- Product description the image and everything else
- Reviews section below description
- If the items stock = 0 then a button labeled “Set stock alert” shall appear besides the “Out of Stock” label
- A heart button that marks a product as favorite and puts it in the Favorite list (this functionality is more detailed below)

Validation: To interact with the Product Detail Page, the system shall validate the following conditions:

- The quantity selector value must be > 0
- The selected quantity must not exceed the available stock quantity. The quantity cannot be bigger than 50.
- If the product stock is 0, The “Add to Basket” button shall be disabled
- The discounted amount in percentages must be > 1
- If invalid quantity input is provided (not natural numbers) the system shall display a message “Invalid quantity selected”

F5.6: - Reviews

Description: The system shall allow users to write and interact with reviews for products through the Reviews Section located on the Product Details Page just below the description. The section will be labeled clearly with a title named “Reviews”. Below this there shall be a form that allows users to enter a rating and a comment. The form shall display a title (“Write Review”) and below it a rating system and a text box with the text of the review. Below this text box there will be a button labeled “Post review” that shall publish the review.

The rating system shall work like this: The system will display a 5 star rating system with half increments. Each star has one rating unit and each star can be divided into 2 selectable areas representing 0.5 stars and 1 star. When a user clicks on a star or half a star the system shall highlight with yellow all the stars from left to right up to the selected star. Then it shall calculate the rating based on how many stars are highlighted (if 1.5 stars are yellow the rating is 1.5). Below this form are the reviews of the product.

Each review shall display:

- The username of the reviewer
- The rating, represented visually using a 5-star system
- The review text
- The date of submission

Users shall also be able to:

- Edit their own reviews. The add review form shall be replaced with the user review that was already posted prior. This review shall display an edit button that shall open the Edit Review form that is almost the same as the add review form with some changes in the text (name of the form, etc). Also in the text box of this form shall be the text body of the review.

- Delete their own reviews. The users submitted review shall also display a delete button that deletes the review. A warning shall be displayed after clicking this button asking confirmation about the deletion.
- Report reviews. Each review shall have a flag button that will notify the admins about review (how the notification is handled is discussed more in Pharmacy Management).

The system shall calculate the average product rating with this formula: (sum of all user reviews/total number of reviews). This value shall be rounded down to the nearest 0.5 increment.

Validation: To submit or interact with reviews the system shall validate the following conditions:

- A user may submit only one review per product, but may edit later.
- The review text must not exceed 500 words.
- Only the author of a review may edit a review
- Only the author or an admin may remove a review

F5.7:- Favorite products

Description: The system shall allow users to mark products as favorite items for easier access.

Each product details page shall contain a heart shaped button that will mark items as favorite. When the user presses this button, the user shall be able to access their favorite products through a Favorites page.

This page is identical to the catalog page but it contains only the Favorites items.
If a product is already marked as favorite the heart icon shall appear filled. Pressing the button again shall remove the product from the favourites list.
If the product is removed from the catalogue, it shall automatically be removed from the users favourites list.

Validation: To manage favorite items the system shall validate the following conditions:

- The favorites list shall display only products that currently exist in the catalogue
- If the favorites list is empty, the system shall display the message: “You have no favorite products.”

F5.8: - Home page

Description: The system shall provide a Home page that acts like the main entry point of the application. The home page has a sidebar with all the categories available. If you click a category the user will be taken to the catalog with that category filter option selected. The home page will also display a topbar that will be available on every page that the user can see. The buttons are the following:

- **Products** that goes to the catalog page
- **Complaints** that goes to the complaints page
- **Cycle Tracker** that goes to the cycle tracker page
- **Admin Dashboard** that goes to the pharmacy management page. This button is only available to users that are admins

- **Login** button that goes to the login page with the login form. This button is replaced by My Account button if the user is logged in
- **Register** button that goes to the Register page with the Register form. This button is replaced by My Account button if the user is logged in
- **My account** button that goes to the account information page This button is replaced by the buttons **Login** and **Register** if the user is not logged in.
- **Cart** button that opens the cart tab
- **Favorites** that goes in the favorites page
- **Home** page that goes to the home page

In the center of the Home page there will be a banner and an overlay with some text. The text should feature:

- The name of the pharmacy in big text
- The motto in a smaller text right below the name
- A quick description about the site with the same text size as the moto below the motto

Validation: The system shall validate the following conditions:

1. The Admin Dashboard button shall be visible only to users that are admins
2. For users not logged the system shall display the My Account button and hide the Login and Register Buttons
3. If a non-authenticated user attempts to press any button besides Products, Home, Login, Register and the cards of the products he/she will be redirected to the Login page
4. If the banner content is unavailable, the system shall display a default banner message.

Feature 6:- Complaints

F6.1:- Complaint creation

Description: Users can send **complaints**, which will be a separate page in the main page of the application, shown by clicking a label on the side of the page with the text “Complaints”. The complaints page has a title and text body that the user can write anything in, UTF-8, and then click the “Send” Button at the bottom of the page to send the complaint to the admins. The complaint also has a “severity” field with some checkbuttons below the title, from 1 to 5. Admins will prioritize complaints with bigger severities. A complaint is automatically assigned to an admin

Validation: To successfully send a complaint, the user needs to provide a *title and text body* for it, i.e. they must not be empty.
If the user is NOT logged in, they cannot access this page, and will be prompted to the log in.

F6.2:- Complaint thread (user + admin)

Description: Complaints are received and can be viewed in another page in the admin dashboard, or by the user in their Account Management section, in the Complaints

section. There the clients will see a list of their complaints, sorted by recent, which will show the title of the complaint and to the right the severity (1-5), and to the right of that the admin handling their complaint. The admin will view their assigned complaints in a similar list on a page in the admin dashboard, accessed from a label titled “Complaints” on their left navigation bar in the dashboard.

A complaint page looks like a *forum thread*, in which the user and the admin can converse in a back and forth until the complaint is closed by the admin. When an admin answers a complaint, or, respectively, closes it, the user receives in their mail the response, or, respectively, that an admin closed their complaint. They also receive a notification that an admin answered their complaint.

Validation: To answer a complaint, the text body needs not be empty. If an admin closes their complaint, the user has the right to **appeal once**: they can reopen the complaint once and send a message in it, which the admin moderating the complaint will receive. If the user is trolling there is no punishment.

If the user is NOT logged in, they cannot access this page, and will be prompted to the log in.

F6.3:- Complaint modification by admin

Description: In the admin dashboard, the admin can close a complaint to mark it as “done”. This will be done by clicking a button to the top right of the thread, where there will be a button entitled “Close complaint”. The admin can also modify the severity, by clicking a button to the left of the close button, called “Change severity”, which will open a dropdown menu where the admin has to select the severity 1-5, then click Save. Save is a button above the close button, which saves the complaint.

Validation: The thread will be marked closed in the database, and no one will be able to reply to it again. The admin has the right to reopen it by clicking on a button on the same place where the “close” button was, entitled “Reopen complaint”

F7:- Cycle tracker

F7.1:- Calculating cycle from data

Description: The application will also have an upside navigation bar link that is called “Cycle Tracker”. When a user (who is logged in) clicks it, a new page opens up.

If the user has never done this before, the page will only show a few labels with their associated fields: First day of your last period (which is a date that the user selects), how much did the last period last, and how many days a cycle lasts for the user on average. There is also an optional field titled “How many days before cycle do the PMS symptoms appear, if any”, which will have below it some checkboxes, “not applicable”, “1-3 days before period”, “4-7 days before period”, “7-14 days before period”. Default is “not applicable”

Below these fields arranged vertically, there will be a button labeled “Calculate Cycle”. Even after pressing it, this part will stay on the bottom of the page always, to recalculate if needed, above the recommended products.

After pressing it, the page updates to show a calendar with 3 months. To the right there is a button titled “Show the next 3 months”, that calculates the cycle for the next 3 months. The days when they have their period are background colored in pink, and the

ones for the ovulation window are colored in light green. The ones with low fertility are colored in yellow, and the PMS days are colored in red. Below the calendar, there will be a legend: a pink square with the label “Period days” to its right, then a light green filled square with the label “Most probable ovulation days” to its right, and a yellow square with “Low fertility days” to its right, and then a red square with “PMS days” to its right.

Validation: If the user is not logged in, they cannot access this page, and will be sent to the log in page. The data have to be in valid format, i.e. the number of days has to be an integer, and the date will be inputted from a date pickup.

F7.2:- Notes for symptoms

Description: Below the information about the cycle, we also have a big textbox where the user can note anything, for example symptoms relating to their cycle to help them keep track of their health better. To the right of this, they have a “+” that, if pressed, will create another note. A note can also be a “TO-DO” note, which means that it will have a checkbox, which ,if clicked, will strike through the whole text, to mark it “done”. This checkbox will be to the left of the note, before the title of the note. A note can be deleted by clicking on the “X” to the upper right corner of a note.

Validation: The notes are written in UTF-8 format, and the user can create at most 4 adjacent notes. After that, the “+” disappears and it will only appear if the user deletes a note.

F7.3:- Recommended related items

Description: Below the notes, we have a checkbox to check if the user wants to get pregnant or not. If yes, then the recommended products at the end of the page will NOT have the contraceptives category too.

At the end of the page, we have some related items (below the calculating data): contraceptives, vitamins, pregnancy tests, in random order. The items are arranged in a horizontal list, showcasing the item thumbnail, its title, and its price, together with a discounted price if the user meets some conditions.

Validation: The products shown to the client below the calculator are in random order, and they will have the option to add them to the basket from there, below the item, which will result in their basket being updated. If the client is in their most probable ovulation window, or in their period (the date of today), give them a 10% discount on those suggested items.